Experiment 6 Day 3: Conversion of Borneol to Camphor

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Chem 2081-007

Introduction

In this experiment, camphor was prepared from the reaction of borneol with sodium hypochlorite. The product was isolated using methylene chloride. The correct product was determined using melting point.

Materials and Methods

To start, the sand bath was plugged into the 10-amp outlet and the thermostat was set to 30. Next, 0.3 grams of the borneol product was added to a long-neck 10 mL round-bottom flask along with 0.3 mL of glacial acetic acid. This solution was then dissolved in 1 mL of acetone, and a boiling chip was added. After this, a reflux apparatus with a distillation column was assembled and placed in a water bath with temperatures between 40-50 degrees Celsius. Once all the reagents had dissolved, 6 mL of aqueous sodium hypochloride was added dropwise through the top opening of the condenser over the span of 35 minutes. The mixture was given a gentle swirl after the addition of each drop of sodium hypochloride. After the last drop of sodium hypochloride, the reaction occurred for 15 minutes. Once the reaction was complete, the mixture was cooled to room temperature and transferred to a centrifuge tube where camphor was extracted with two portions of 2 mL of methylene chloride (DCM), 4 mL in total. The two layers were then separated into different tubes, and the organic layer was put back into the original centrifuge tube. Following this, the methylene chloride layers were neutralized with 2 mL of saturated sodium bicarbonate. The two present layers were separated into different tubes again, and the organic layer was put back into the original centrifuge tube. Next, the organic solution was washed sequentially with 2 mL of 5% sodium bisulfite and 2 mL of water. The two layers were separated into two different tubes, and the organic solution was dried over anhydrous sodium sulfate. Once the drying was complete, the DCM was evaporated over a water bath that

did not exceed 50 degrees Celsius, and the collected crystals were weighed. Following this, all products were disposed of in the proper container, and all glassware was washed and left to air dry.

Results

Theoretical Yield (g)	0.3
Actual Yield (g)	0.162
Percent Yield (%)	54

Table 1. Theoretical Yield, Actual Yield, and Percent Yield of Camphor

$$0.3\ g\ of\ C10H18O * \frac{154.25\ g}{1\ mol\ C10H18O} * \frac{1\ mol\ C10H18O}{1\ mol\ C10H16O} * \frac{1\ mol\ C10H16O}{152.23\ g\ C10H16O}$$

$$= 0.30\ g\ of\ C10H16O\ (Camphor)$$

$$\frac{0.162}{0.3} * 100\% = 54\%$$

Discussion

In the experiment, 0.3 grams of borneol reacted with sodium hypochlorite and glacial acetic acid to form camphor. Borneol and glacial acetic acid acted as the reagents and the sodium hypochlorite was the oxidizing agent. The reaction occurred in 40ºC to 50ºC to yield maximum product. The product was isolated using a methylene chloride wash and the crystals were collected after the evaporation of DCM. The correct product was obtained and determined using density and melting point. The organic layer in the reaction after the addition of methylene chloride was the bottom layer because it has a density of 1.33 g/mL, and the reaction mixtures density is 1 g/mL. After the addition of water and 5% sodium bisulfite the organic layer stayed as

the bottom layer because the product is more dense than water allowing it to sink to the bottom of the centrifuge tube. Another way to determine the correct product was collected is by using melting point. The last step of the experiment involved the evaporation of DCM using a water bath. This left behind crystals that had a melting point of approximately 175ºC which matched the literature melting point of 175ºC-177ºC for camphor. Two possible sources of error that could have affected the results of the experiment were the leaving behind of the drying agent which would have affected the final weight of the product and overheating the reaction mixture causing loss of product due to the reagents and product being highly volatile. The most likely source of error was overheating the reaction mixture which would have caused loss of product. Both borneol and camphor are highly volatile, meaning they evaporate readily at lower temperatures, so the water bath had to be kept at a temperature between 40ºC and 50ºC. This was difficult to do with the set-up, so one way to fix this problem would be to use a thermal immersion circulator which is a device that keeps the water at a desired constant temperature. This would have allowed the water to stay between 40ºC and 50ºC and kept the reagents and product from evaporating. This would have allowed for maximum yield of camphor.

References

. Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry 2 Lab Manual [lab manual]. Auburn (Al): Auburn University.

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Exp. No. 6 Day 3	Experiment/Subject Conversion of Borneol to Camphor	Date 11-16-23	
Name Terra Black	Lab Partner	Locker/ Desk No. Hood H	Course & Section No. 2081 007

Procedure	Observation
1. In a long-neck 10 mL round-bottom flask, add racemic 0.3g. of borneol product with 0.3 mL of glacial acetic acid. Dissolve the racemic mixture in 1mL of acetone. Add a boiling chip.	1. Used borneol from 3 different sources because the last experiment did not yield enough product.
2. Assemble a reflux with the distillation column and place apparatus in a water bath with temperature between 40-50°C. Make sure the temperature does not deviate from the specified range. If the borneol does not dissolve at this temperature, add 0.5 mL of acetone to the mixture until all borneol goes into the solution.	2. Turned sand bath down to 30 to keep temperature in the specified range. Borneol dissolved quickly. Took 15-20 minutes to get temperature of water bath stabilized.
3. Once all reagents are dissolved, add 6 mL of aqueous ~~solution~~ sodium hypochloride solution (bleach). The addition should be done through the top opening of the condenser and should be paad dropwise over 35 minutes. After each addition of the hypochloride, the mixture should be given a gentle swirl to ensure all chemicals are mixed thoroughly. The reaction must be allowed to continue for 15 minutes after the last bleach addition.	3. Added 1 mL of bleach over the span of 2-3 minutes. Reaction occurred for 15 minutes at a slight reflux. Wrapped the condenser in wet paper towels to keep product from evaporating out.
4. Let the mixture cool to room temperature and extract the camphor with two portions (2 mL each) of methylene chloride (or DCM). Use a centrifuge tube to make the extraction more efficient.	4. Tiny bit spilled when tube holder fell off the "monkey bars". DCM went to the bottom layer.

Signature	Date	Witness/TA	Date

THE HAYDEN-McNEIL STUDENT LAB NOTEBOOK

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Exp. No. 6 Day 3	Experiment/Subject: Conversion of Borneol to Camphor	Date: 11-16-23	
Name: Terra Black	Lab Partner	Locker/Desk No. 1602 H	Course & Section No. 2081 007

Procedures Continued

5. Neutralize the combined methylene chloride layers with ~2mL of saturated sodium bicarbonate. Keep adding bicarbonate solution if necessary. The presence of acidic residuals will be determined by the production of CO_2 bubbles after every addition of the bicarbonate solution.
6. Wash the organic solution sequentially with 2 mL of 5% sodium bisulfite and with 2 mL of water.
7. Dry the organic solution over anhydrous sodium sulfate. The solution is considered dry when freely rolling crystals of the drying agent are observed.
8. Evaporate the DCM over a water bath with temp. not exceeding 50°C. Weigh the collected crystals and determine your percent yield. Compare melting point w/ that published in literature.
9. Drying agent can be washed down the drain with plenty of water. and residual DCM and all other organic products are disposed of in the appropriate containers in the fume hood.

Observations Continued

5. saturated sodium bicarbonate was the top layer. Some bubbles occurred.
6. Organic/Product layer was the bottom layer.
7. Used approximately 2 small spatulas of drying agent. Made sure to keep all drying agent in tube when transferring product to watchglass
8. Did not take melting point.
9. 50.585 = watchglass before
   50.747 = after w/ product
   0.162 g. of camphor = actual yield

Signature	Date	Witness/TA	Date

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Lab Performance

Exp 6 D3

Terra Black | 007 | 402 | H

Name: | *Section:* | *Room:* | *Hood:*

Wearing personal protection equipment—this evaluation records your consistency with wearing the appropriate clothes for lab (long pants, closed toe shoes, and tops with sleeves) as well as the required laboratory safety equipment of apron, goggles, and gloves. ○ ○ ○ ○ ~~○ (5pts)~~

Safe working habits—as the GTAs do the lab rotations, they will record your work ethic and whether you follow our safety rules and regulations concerning tied hair, tied apron, safety goggles on at all times, and no food and drinks. ○ ○ ○ ○ ○ (5pts)

Following our laboratory safety technique work procedures—evaluation of operation of our laboratory equipment—water sinks in hood (1pt), vacuum and air lines (1pt), sash position (1pt), and heating elements (1pt). It is important that after every experiment you shut down the hood and make it ready for the section after yours (5pts). ○(1pt) ○(1pt) ○(1pt) ○ (1pt) ○ (5pts)

Glassware operation—safe handling of our glassware equipment. This means cleaning (1pt), maintaining the drawer inventory (5pts), and proper experimental setup (5pts). ○(1pt) ○ (5pts) ○ (5pts)

Working independently or equal partner participation during the entire lab. ○ ○ ○ ○ ○ (5pts)

Good work behavior—you work in a dynamic environment; therefore, creating distractions through noise and disobedience will not be tolerated. No cell phone use, listening to loud music, or working/distracting at a non-assigned workspace. ○ ○ ○ ○ ○ (5pts)

Maintain clean and organized personal space. ○ ○ ○ ○ ○ (5pts)

GTA notes:

GTA signature: _____ *Final Performance Grade:* B